

EduShell Review – Easy Explanation Guide

Opening statement: “EduShell is my mini Unix-like shell project. I developed it step by step and implemented command execution, pipelines, redirection, memory monitoring, signals, and job control.”

1. Open EduShell

Command:
``cd ~/EduShell``
``./EduShell``

Say: “This launches my custom shell.”

2. Help command

Command:
``help``

Say: “This displays the built-in commands supported by EduShell.”

3. Basic execution

Command:
``echo Hello``

Say: “This shows command execution using processes and system calls such as fork and exec.”

4. Pipeline

Command:
``ls | wc -l``

Say: “The output of ls is passed as input to wc using a pipe.”

5. Write redirection

Command:
``echo Hello > demo.txt``
``cat demo.txt``

Say: “The > operator writes command output into a file.”

6. Append redirection

Command:
``echo World >> demo.txt``
``cat demo.txt``

Say: “The >> operator appends data without overwriting the existing file.”

7. Virtual memory

Command:
``memstat``

Say: “This displays memory information such as VmSize and VmRSS for the shell process.”

8. Background job

Command:
``sleep 20 &``
``jobs``

Say: “The ampersand runs the command in the background, and jobs lists active jobs.”

9. Stop a foreground job

Command:
``sleep 30` then press Enter, then press Ctrl+Z`

Say: “Ctrl+Z suspends the foreground process and records it as a stopped job.”

10. Resume in background

Command: <code>`jobs`</code> <code>`bg %N`</code> <code>`jobs`</code>	Say: “bg resumes the stopped job in the background. N is the job number shown by jobs.”
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11. Bring to foreground

Command: <code>`fg %N`</code>	Say: “fg brings the selected job back to the foreground.”
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12. Exit

Command: <code>`exit`</code>	Say: “This terminates the shell cleanly.”
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Closing statement

“Through this project, I learned how processes, system calls, pipes, file descriptors, signals, virtual memory, and job control work together to build a shell.”

Important reminder

For Ctrl+Z: type **sleep 30**, press Enter, wait for it to run, and then physically press **Ctrl + Z**. Do not type ^Z as text. Use the job number displayed by jobs when running bg or fg.